



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

AIOT

28-05-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	C	A	B	B	A	B	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	D	C	B	D	C	D	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	70	5	2	4	50	6	5	3	2400	4

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	B	D	B	B	A	B	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	D	D	D	A	C	D	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3	63	2	12	7	495	2	5	7	6

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	C	A	C	A	B	B	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	D	D	B	D	B	B	C	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	14	25	5	4	4	2	64	3	1	8

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (C)**

$$\frac{v_1 \sin \theta_1}{g} - \frac{v_2 \sin \theta_2}{g} = T$$

$$\frac{v_1^2 \sin^2 \theta_1}{2g} - \frac{v_2^2 \sin^2 \theta_2}{2g} = H$$

Dividing both

$$\Rightarrow v_1 \sin \theta_1 + v_2 \sin \theta_2 = \frac{2H}{T}$$

$$\Rightarrow m(v_1 \sin \theta_1 + v_2 \sin \theta_2) = \frac{2mH}{T}$$

2. **Ans (A)**

Rays left of O

$$\sin 60^\circ = \sin r \times 2$$

$$\frac{\sqrt{3}}{2} = 2 \sin r$$

$$\therefore r = \sin^{-1} \frac{\sqrt{3}}{4}$$

$$\frac{\sqrt{3}}{4} = \sin 25^\circ$$

$$\phi = 180 - [90 - r + \theta_c]$$

$$\text{Where } \theta_c = \sin^{-1} \frac{1}{2} = 30^\circ$$

$$\phi = 180 - [90 - 25^\circ + 30^\circ] = 85^\circ$$

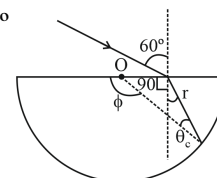
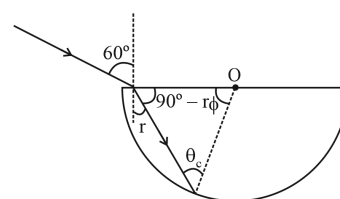
Rays right of O

$$180^\circ - \phi + 90^\circ + r + \theta_c = 180^\circ$$

$$-\phi + 90^\circ + 25^\circ + 30^\circ = 0$$

$$\phi = 145^\circ$$

$$\therefore 85^\circ < \phi < 145^\circ$$



3. **Ans (C)**

Flux due to charge at center is $\frac{Q}{2\epsilon_0}$

Flux due to $-Q$ is $\frac{Q}{2\epsilon_0}(1 - \cos 37^\circ) = \frac{Q}{10\epsilon_0}$

$$\therefore \text{Total flux} = \frac{Q}{2\epsilon_0} + \frac{Q}{10\epsilon_0} = \frac{3Q}{5\epsilon_0}$$

4. **Ans (A)**

$$\frac{1}{\sqrt{LC}} = 2\pi f$$

$$f = 100 \times 10^6 \text{ Hz}$$

solving $L = 0.25 \mu\text{H}$

$$\therefore [0.2 + (0.1) \sin \theta] = 0.25$$

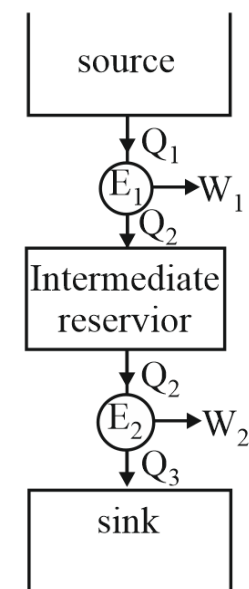
$$\therefore \sin \theta = 0.5$$

$$\Rightarrow \theta = 30^\circ$$

5. **Ans (B)**

Minima will be detected when path differences is $\frac{\lambda}{2} = 1\text{m}$ and that is when S_2 has moved distance 3 meter.

6. **Ans (B)**



$$n_1 = 1 - \frac{Q_2}{Q_1} = \frac{1}{5} \quad \dots(1)$$

$$n_2 = 1 - \frac{Q_3}{Q_2} = \frac{2}{5} \quad \dots(2)$$

From (1) and (2)

$$n = 1 - \frac{Q_3}{Q_1} = \frac{13}{25}$$

7. **Ans (A)**

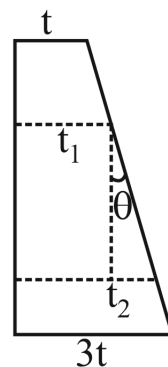
Shift in fringe pattern

$$= \frac{t_2(\mu - 1)}{\lambda} \beta - \frac{t_1(\mu - 1)}{\lambda} \beta$$

$$= \frac{(t_2 - t_1)(\mu - 1)\beta}{\lambda}$$

$$= \frac{d \tan \theta (\mu - 1)\beta}{\lambda} \quad (\text{constant}) ;$$

$$= \frac{2t(\mu - 1)D}{3d}$$



8. **Ans (B)**

$$\phi = \phi_0 r^4$$

$$\phi = \phi_0 r^4 = E 4\pi r^2$$

$$\Rightarrow E = \frac{\phi_0}{4\pi} r^2$$

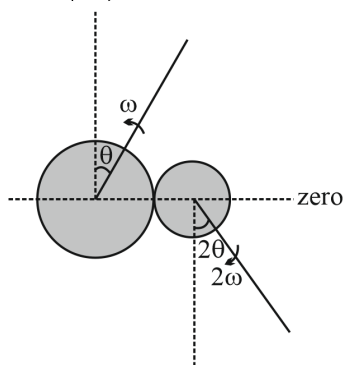
$$dU = \left(\frac{1}{2} \epsilon_0 E^2 \right) 4\pi r^2 dr = \frac{\epsilon_0}{2} \frac{\phi_0^2}{16\pi^2} r^4 4\pi r^2 dr$$

$$= \left(\frac{\epsilon_0 \phi_0^2}{8\pi} \right) r^6 dr$$

$$U = \left(\frac{\epsilon_0 \phi_0^2}{8\pi} \right) \int_0^r r^6 dr$$

$$= \frac{\epsilon_0 \phi_0^2 r^7}{56\pi}$$

9. Ans (B)



$$mg \frac{\ell}{2} \cos \theta - mg \frac{\ell}{2} \cos 2\theta + \frac{1}{2} \left(\frac{m\ell^2}{3} \right) \omega^2 + \frac{1}{2} \left(\frac{m\ell^2}{3} \right) (2\omega)^2 = \text{constant}$$

$$mg \frac{\ell}{2} \left(1 - \frac{\theta^2}{2} - 1 + 2\theta^2 \right) + \frac{1}{2} \frac{m\ell^2}{3} \omega^2 (1 + 4) = \text{constant}$$

$$\frac{5}{6} m\ell^2 \omega^2 + \frac{3}{4} mg\ell \theta^2 = \text{constant}$$

$$\frac{m\ell}{2} \left(\frac{5}{3} \omega^2 \ell + \frac{3g\theta^2}{2} \right) = \text{constant}$$

$$\frac{5}{3} \omega^2 \ell + \frac{3}{2} g\theta^2 = \text{constant}$$

$$\frac{5}{3} \ell 2\omega \frac{d\omega}{d\theta} + \frac{3}{2} g 2\theta = 0$$

$$\frac{10}{3} \ell \alpha + 3g\theta = 0$$

(angular SHM equation)

$$(\text{angular frequency})^2 = \frac{9g}{10\ell}$$

$$T = 2\pi \sqrt{\frac{10\ell}{9g}}$$

10. Ans (D)

$$X_L = 100 \times \frac{\sqrt{3}}{10} = 10\sqrt{3}$$

and $R = 10\Omega$

$$\therefore \tan \phi_1 = \frac{X_L}{R} = \sqrt{3} \Rightarrow \phi_1 = 60^\circ$$

$$X_C = \frac{1}{\omega C} = \frac{2 \times 10^4}{\sqrt{3}}$$

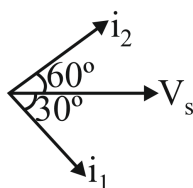
and $R = 20 \times 10^3$

$$\therefore \tan \phi_2 = \frac{X_C}{R}$$

$$\Rightarrow \phi_2 = 30^\circ$$

So,

$$\Delta \phi = 90^\circ$$



11. Ans (A)

$$Y = A \sin (\omega t - \vec{k} \cdot \vec{r})$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{k} \cdot \vec{r} = (3x + 2y - z)\pi$$

$$\vec{k} = (3\hat{i} + 2\hat{j} - \hat{k})\pi$$

$$|\vec{k}| = \pi\sqrt{9 + 4 + 1} = \pi\sqrt{14}$$

$$|\vec{k}| = \frac{2\pi}{\lambda}$$

$$\Rightarrow \lambda = \frac{2\pi}{|\vec{k}|} = \frac{2\pi}{\pi\sqrt{14}} = \sqrt{\frac{2}{7}} \text{ m}$$

Unit vector in the direction of wave

$$\text{propagation} = \frac{3\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{14}}$$

12. Ans (A)

$$\text{Rotating torque } (\tau_1) = \int_0^h \rho g y \, bdy (h - y)$$

$$= \rho g b \frac{h^3}{6}$$

$$\text{Opposing } (\tau_2) \text{ torque} = \rho g h \frac{\ell}{2} \times \ell \times b$$

gate will open if $\tau_1 > \tau_2$

$$\therefore h > \ell\sqrt{3}$$

$$\frac{h}{\ell} > \sqrt{3}$$

Hence $x = 1.73$

13. Ans (D)

The electron ejected with maximum speed $v_{\max}$ are stopped by electric field $E = 0.9 \text{ N/C}$ after travelling a distance $d = 1.5 \text{ m}$

$$\therefore \frac{1}{2} m v_{\max}^2 = eEd = 1.35 \text{ eV}$$

$$\text{The energy of incident photon} = \frac{1240}{400} = 3.1 \text{ eV}$$

From equation of photo electric effect

$$\frac{1}{2} m v_{\max}^2 = h\nu - \phi_0$$

$$\therefore \phi_0 = 3.1 - 1.35 = 1.75 \text{ eV.}$$

14. Ans (C)

Let charge on innermost shell be x , on middle shell be y and on outermost shell be $3 - x$.
Potential of middle shell be zero since it is earthed

$$\frac{Kx}{0.6} + \frac{Ky}{0.6} + \frac{K(3-x)}{1.2} = 0$$

$$\Rightarrow x + 2y + 3 = 0$$

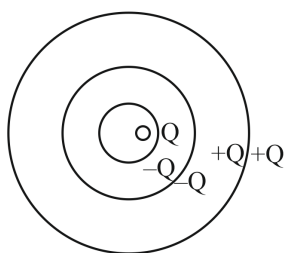
Potentials of inner and outer shells are same.

$$\frac{Kx}{0.4} + \frac{Ky}{0.6} + \frac{K(3-x)}{1.2} = \frac{Kx}{1.2} + \frac{Ky}{1.2} + \frac{K(3-x)}{1.2}$$

$$\Rightarrow 2x + y = 0$$

Solving $x = 1\mu\text{C}$ $y = -2\mu\text{C}$

Charge distribution :



15. Ans (B)

$$a = -10\hat{k} \text{ m/s}^2$$

$$\text{Initial position vector } \vec{r}_1 = 2\hat{i} + 3\hat{j}$$

$$\text{Final position vector } \vec{r}_2 = \vec{r}_1 + \vec{u}t + \frac{1}{2}\vec{a}t^2$$

$$\vec{r}_2 = 2\hat{i} + 3\hat{j} + (\hat{i} + 2\hat{j} + 10\hat{k})t - 5t^2\hat{k}$$

$$\vec{r}_2 = (2+t)\hat{i} + (3+2t)\hat{j} + (10t-5t^2)\hat{k}$$

$$\text{When particle hits ground } 10t - 5t^2 = 0$$

$$t = 2 \text{ sec}$$

Coordinates when it hits ground is (4, 7, 0)

16. Ans (D)

Frequency of revolution of particle

$$f = \frac{qB}{2\pi m} = \frac{2 \times 10^2 \times 7}{2\pi} = \frac{7 \times 10^2}{\pi} \text{ Hz}$$

Pitch of the helical path

$$P = TV \cos\theta = \frac{1}{f} V \cdot \left(\frac{\vec{V} \cdot \vec{B}}{VB} \right)$$

$$= \frac{1}{f} \left(\frac{\vec{V} \cdot \vec{B}}{B} \right) = \frac{10\pi}{49} \text{ cm}$$

17. Ans (C)

Since AB is rectangular hyperbola

$$U\rho = \text{constant}$$

$$\frac{nfRT}{2} \frac{m}{V} = \text{constant}$$

$$\frac{T}{V} = \text{constant}$$

$$P = \text{constant}$$

$$\text{For A: } 5U_0 = n \frac{3R}{2} T_A$$

$$\frac{10}{3} U_0 = nRT_A$$

$$\text{For B: } 2U_0 = n \frac{3R}{2} T_B$$

$$\frac{4}{3} U_0 = nRT_B$$

$$W_{AB} = nR(T_B - T_A) = \frac{4U_0}{3} - \frac{10U_0}{3} = -2U_0$$

$$U_A\rho_A = U_B\rho_B$$

$$5U_0 2\rho_0 = 2U_0 \rho_B$$

$$\rho_B = 5\rho_0$$

Process CA is isothermal

$$W_{CA} = nRT_A \ln\left(\frac{V_A}{V_C}\right)$$

$$= nRT_A \ln\left(\frac{\rho_C}{\rho_A}\right) = \frac{10}{3} U_0 \ln \frac{5}{2}$$

In cyclic process

$$Q_{\text{net}} = W_{AB} + W_{BC} + W_{CA}$$

$$\therefore [W_{BC} = 0]$$

$$= \left[\frac{10}{3} \ln(2.5) - 2 \right] U_0$$

18. Ans (D)

$$i = \frac{dQ}{dt} = -K_0 \left(1 + \frac{x}{L} \right) S \frac{d\theta}{dx}$$

i = heat current

$$i \int_0^L \frac{dx}{\left(1 + \frac{x}{L} \right)} = -K_0 S \int_{3\theta}^{\theta} d\theta$$

$$iL \int_0^L \frac{dx}{L+x} = 2K_0 S\theta$$

$$iL \ln 2 = 2K_0 S\theta$$

$$i = \frac{2K_0 S\theta}{L \ln 2}$$

19. **Ans (C)**

Before voltmeter connected

$$V_1 = \frac{4}{4+6} \times 20 = 8V$$

After voltmeter connected

$$V_2 = \frac{3}{3+6} \times 20 = \frac{20}{3} V$$

$$\text{Percentage error} = \frac{8 - \frac{20}{3}}{8} \times 100 = 16.7\%$$

20. **Ans (B)**

When S_1 closed then current is from A to O hence torque on rod AB is clockwise.

When S_2 closed then current is from B to O hence torque on rod AB is anticlockwise.

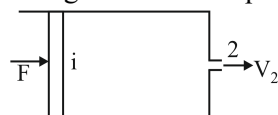
When both are closed then torque is zero.

PART-1 : PHYSICS

SECTION-II

1. **Ans (70)**

Using Bernoulli's equation

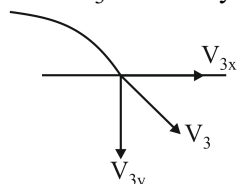


$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2 \quad \because [V_1 \ll V_2]$$

$$P_0 + \frac{F}{A} = P_0 + \frac{1}{2}\rho V_2^2$$

$$V_2 = \sqrt{\frac{2F}{\rho A}} = \sqrt{50} \text{ m/s}$$

Here V_3 = velocity when it hits ground



$$V_{3x} = \sqrt{\frac{2F}{\rho A}}$$

$$V_{3y} = \sqrt{2gh}$$

$$V_3 = \sqrt{V_{3x}^2 + V_{3y}^2} = \sqrt{\frac{2(F + \rho ghA)}{\rho A}} = 8.366$$

2. **Ans (5)**

Let $h \propto d^x Y^y g^z$

on solving $h \propto \frac{Y}{dg}$

$$\text{Now } \frac{dh}{h} \times 100$$

$$= \pm \left[\frac{dY}{Y} \times 100 + \left(\frac{d(d)}{d} \times 100 \right) + \left(\frac{dg}{g} \times 100 \right) \right]$$

$$= \pm 5\%$$

3. **Ans (2)**

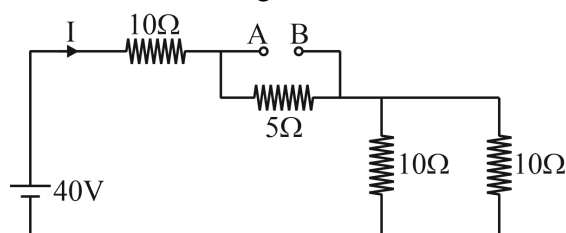
Forming WET equation,

$$Fd = \frac{1}{2} Kd^2 + \frac{1}{2} (2K)d^2$$

$$\frac{2F}{3K} = d.$$

4. **Ans (4)**

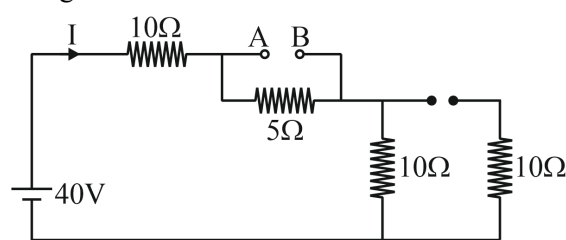
At the instant just before $t = 0$, the equivalent circuit is shown in figure below



$$I = \frac{40}{10 + 5 + \frac{10 \times 10}{10 + 10}} = 2A$$

$$\therefore V_C = V_A - V_B = 5 \times 2 = 10V$$

At time $t \rightarrow \infty$, the equivalent circuit is shown in figure below :



$$I = \frac{40}{10 + 5 + 10} = \frac{40}{25}$$

$$\therefore V_C = V_A - V_B = 5 \times \frac{40}{25} = 8V$$

The equivalent resistance across the capacitor is given by

$$R_{eq} = \frac{(10 + 10)5}{20 + 5} = 4\Omega$$

$\therefore \tau$ = Time constant of the discharge circuit

$$= CR_{eq}$$

$$= \left(\frac{1}{16} \right) (4) = \frac{1}{4} \text{ sec}$$

$$\text{Now, } V_C = A + Be^{-kt}$$

$$\text{At } t = 0, V_C = 10V \text{ or, } 10 = A + B$$

$$\text{At } t \rightarrow \infty, V_C = 8V \text{ or, } 8 = A + 0$$

$$\text{or, } A = 8V$$

$$B = 2V$$

$$\text{Also, } k = \frac{1}{\tau} = \frac{1}{1/4} = 4 \text{ s}^{-1}$$

5. Ans (50)

$$\mu = Kx = \frac{dM}{dx}$$

$$\int_0^M dM = \int_0^{\ell} Kx dx = \text{and } K = \frac{2M}{\ell^2}$$

$$V = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{F}{Kx}} = \frac{dx}{dt}$$

$$\frac{dx}{dt} = \sqrt{\frac{F \ell^2}{2Mx}}$$

$$\int_0^{\ell} \sqrt{x} dx = \sqrt{\frac{F \ell^2}{2M}} \int_0^t dt$$

$$\therefore t = \sqrt{\frac{8M\ell}{9F}} = \sqrt{\frac{8 \times 0.05 \times 0.9}{9 \times 16}} = 0.05s$$

6. Ans (6)

$$\frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow -\frac{df}{f^2} = \left(\frac{1}{R_1} - \frac{1}{R_2} \right) dn$$

(differentiating both sides)

$$\Rightarrow -\frac{df}{f} (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \left(\frac{1}{R_1} - \frac{1}{R_2} \right) dn$$

$$\Rightarrow -df = f \frac{dn}{n-1}$$

but $dn = n_v - n_R$

$$\therefore -df = \frac{n_v - n_R}{n-1} f$$

$$\Rightarrow df = -\omega f$$

$$\Rightarrow f_v - f_R = -\omega f$$

$$= -0.03 \times 20 = -0.6 \text{ cm}$$

7. Ans (5)

$$A = 4\pi r^2 = 4\pi \left(\frac{10}{\sqrt{\pi}} \right)^2 = 400 \text{ m}^2$$

Intensity $I = \epsilon_0 C E_{rms}^2$

and also $I = \frac{P}{4\pi r^2} \eta$

$$\epsilon_0 C E_{rms}^2 = \frac{P \eta}{4\pi r^2}$$

$$E_{rms} = \sqrt{\frac{60 \times 8.85}{100 \times 8.85 \times 10^{-12} \times 3 \times 10^8 \times 400}}$$

$$E_{rms} = \sqrt{5} \text{ v/m}$$

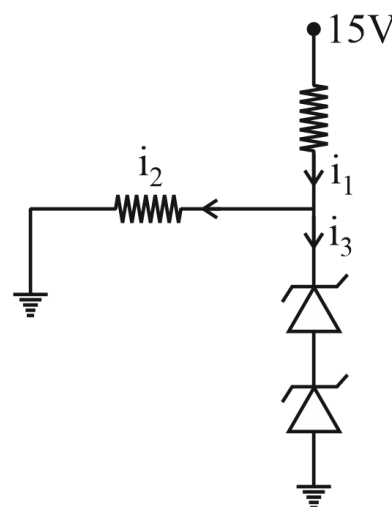
$$= 2.23 \text{ to } 2.24 \text{ v/m}$$

8. Ans (3)

N_0 number of nuclei corresponds to 2ml.
9 : 00 am to 10 : 30 am time difference is equal to 1 half life. Number of nuclei remaining at 10 : 30 am is $\frac{N_0}{2}$.
9 : 00 am to 11 : 15 am time difference is equal to 1.5 half life.
 $\frac{N_0}{2}$ quantity of dose required at 11 : 15 am
 $\frac{N_0}{2} = N_i e^{-\lambda t}$ Put $t = 1.5 \frac{\ln 2}{\lambda}$
 $\therefore N_i = \sqrt{2} N_0$
 $\sqrt{2} N_0$ nuclei required to be prepared at 9 : 00 am which corresponds to $2\sqrt{2}$ ml dosage
 $\therefore$ dose required = 2.82 ml

9. Ans (2400)

Potential drop across R_2 is 4.8V



$$\therefore i_2 = \frac{4.8}{300} = 16 \text{ mA}$$

Potential drop across R_1 is $15 - 4.8 = 10.2$

$$\therefore i_1 = \frac{10.2}{600} = 17 \text{ mA}$$

Current through Zener diodes is 1mA power across each Zener diode

$$P = Vi$$

$$= 2.4 \times 1$$

$$= 2.4 \text{ mJ}$$

10. **Ans (4)**

For sun-earth system

$$T = 2\pi \sqrt{\frac{R^3}{GM_s}} = 1 \text{ yr} \quad \dots(1)$$

For binary star system

$$T = 2\pi \sqrt{\frac{r^3}{G(m_1 + m_2)}} = 2\pi \sqrt{\frac{r^3}{G\left(\frac{M_s}{4} + \frac{3M_s}{4}\right)}}$$

$$= 8 \text{ yr.}$$

$$2\pi \sqrt{\frac{r^3}{GM_s}} = 8 \text{ yr.} \quad \dots(2)$$

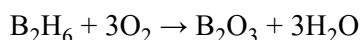
Solving (1) & (2)

$$\frac{R}{r} = 0.25$$

PART-2 : CHEMISTRY

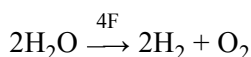
SECTION-I

1. **Ans (D)**



According to balanced equation 27.6 g B_2H_6 i.e. 1 mole B_2H_6 requires 3 mole of O_2 .

Now this oxygen is produced by electrolysis of water as follows



As 1 mole O_2 is produced by 4 F charge

So 3 mole O_2 will be produced by 12 F charge

So now on applying

$$Q = It$$

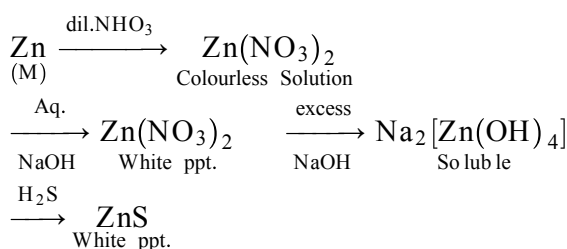
$$12 \times 96500\text{C} = 100 \times t(\text{s})$$

$$t = \frac{12 \times 96500}{100 \times 3600} \text{ hours}$$

$$t = 3.2 \text{ hours}$$

2. **Ans (C)**

Here the metal is Zn and the reactions as follows



3. **Ans (B)**

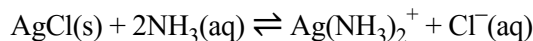


$$0.1 \text{ mol} \quad 28.7 \text{ g}$$

$$\text{No. of moles of AgCl} = \frac{28.7}{143.5} \approx 0.2 \text{ mol}$$

Hence it required 2 mole free Cl atom.

4. **Ans (D)**



$$K_1 = \frac{0-2x}{1.6 \times 10^{-10} \times 10^7} = 1.6 \times 10^{-3}$$

$$K_1 = \frac{[\text{Ag(NH}_3)_2^+][\text{Cl}^-]}{[\text{NH}_3]^2}$$

$$\frac{[\text{Cl}^-]}{[\text{CN}^-]} = \frac{1.6 \times 10^{-3}}{2.5 \times 10^{-9}} = 6.4 \times 10^5$$

$$K_1 = \frac{x^2}{(1-2x)^2} = 1.6 \times 10^{-3}$$

$$\frac{x}{(1-2x)} = 0.04$$

$$x = 0.037$$

$$[\text{CN}^-] = \frac{0.037}{6.4 \times 10^5} = 5.78 \times 10^{-8}$$

5. **Ans (B)**

Effective number of remaining

$$\text{Cl}^- = 4 - \left[2 \times \frac{1}{8} + \frac{1}{2} \right] = \frac{13}{4}$$

Effective number of $\text{Na}^+ = 4$

$$\text{Packing fraction} = \frac{\text{Occupied volume}}{\text{Total volume}}$$

$$= \frac{\frac{13}{4} \times \frac{4}{3} \pi r_-^3 + 4 \times \frac{4}{3} \pi r_+^3}{8(r_+ + r_-)^3}$$

$$\frac{\frac{13}{3} \pi r_-^3 + \frac{16}{3} \pi r_+^3}{8(r_+ + r_-)^3}$$

6. **Ans (B)**

Total charge passed in 30 sec.

$$= \frac{1}{2} \times 100 \times 10 + 100 \times 10 + 100 \times 10 +$$

$$\frac{1}{2} \times 100 \times 10$$

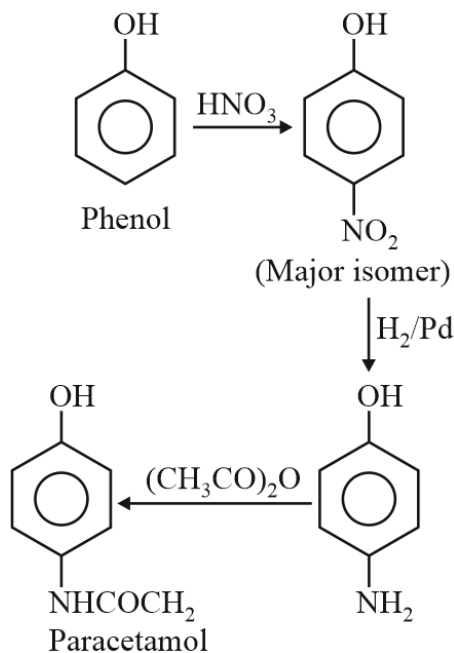
$$= 3 \times 100 \times 10 \text{ mA} \times \text{s}$$

$$= 3000 \times 10^{-3} \text{ amp s} = 3\text{C}$$

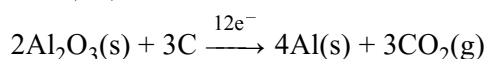
$$W = ZQ$$

$$\text{So, } Z = \frac{W}{Q} = \frac{m}{3}$$

7. Ans (A)



8. Ans (B)



$$\begin{aligned}\Delta G^\circ &= 3\Delta G_f^\circ(\text{CO}_2) - 2\Delta G_f^\circ(\text{Al}_2\text{O}_3) \\ &= -3 \times 394 - 2(-1520) \\ &= -1182 + 3040 = +1858 \text{ kJ}\end{aligned}$$

$$\Delta G^\circ = -nFE^\circ$$

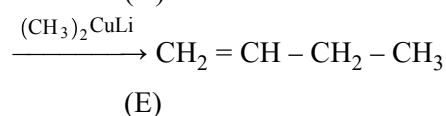
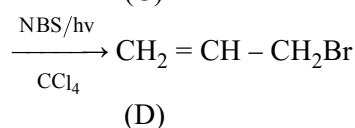
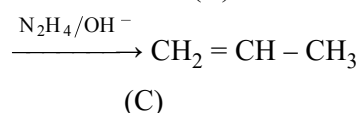
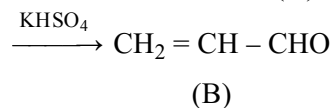
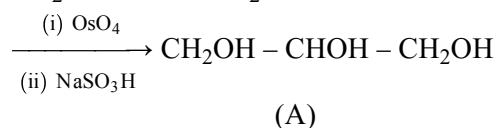
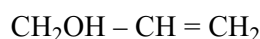
$$1858 \times 1000 = -12 \times 96500 \times E^\circ$$

$$\therefore E^\circ = -1.60 \text{ V}$$

Thus, voltage requires = 1.60 V

9. Ans (B)

Here the final product E is Butene and the reaction sequence is as follows



10. Ans (B)

	List-I		List-II
P.	α -D and β -D glucose	2.	Anomers
Q.	D-glucose and D-galactose	3.	Epimers
R.	Erythrose and threose	4.	Diastereomers
S.	D-glyceraldehyde and L-glyceraldehyde	1.	Enantiomers

11. Ans (A)

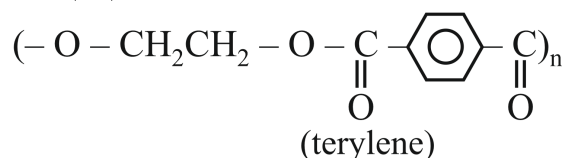
As when temperature increase, vapour pressure increases so C and D options get rejected.

$$\Delta T_f = K_f \times m$$

$$273 - T'_f = 2 \times \frac{34.5}{0.5}$$

$$\text{so } T'_f = 270\text{K}$$

12. Ans (C)



13. Ans (D)

A. Both SbF_4^- and SF_4 have sea-saw structure

B. IOF_5 has sp^3d^2 hybridisation

C. Both SOF_4 and XeO_3F_2 have trigonal bipyramidal structure where double bond is present at equatorial position.

14. Ans (D)

A. Borosilicate glasses contains B and sometimes Al. It can withstand heat changes without cracking. Such glasses are widely used for making good quality laboratory equipments. This glass has low coefficient of expansion.

B. Calcium-alkali-silicate glass is made by heating alkali metal carbonate, CaCO_3 and SiO_2 . This glass is used as domestic glass for windows and bottles.

C. Most of CaO can be replaced by PbO giving lead glass, which has higher reflective index and is used for making optical parts (lenses) and glass ornaments.

D. Soda glass is obtained by heating Na_2CO_3 , CaCO_3 and SiO_2 . It is used for cheap laboratory glass wares.

16. **Ans (A)**

$$k_{\text{eff}} = k_1 + k_2$$

$$(P)_t = 1 \times e^{-kt}$$

$$= e^{-3 \times 10^{-2} \times 33.3}$$

$$= e^{-0.99} = \frac{1}{e}$$

17. **Ans (C)**

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$\lambda_1 = \frac{h}{\sqrt{2me(100)}} = \frac{h}{10\sqrt{2me}} = \frac{k}{10}$$

$$k = \frac{h}{\sqrt{2me}}$$

$$\lambda_2 = \frac{h}{\sqrt{2me(81)}} = \frac{k}{9}$$

$$\lambda_3 = \frac{h}{\sqrt{2me(49)}} = \frac{k}{7}$$

$$\frac{\lambda_3 - \lambda_2}{\lambda_1} = \frac{\frac{k}{7} - \frac{k}{9}}{\frac{k}{10}} = \frac{\left(\frac{2k}{63}\right)}{\left(\frac{k}{10}\right)} = \frac{20}{63}$$

18. **Ans (D)**

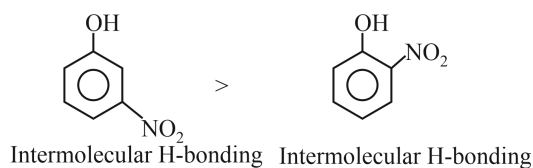
Here option D is incorrect while rest are correct as follows

(A) $T_2 > D_2 > H_2$ [BP $\propto$ mol. wt.]

(B) n-pentane > neo-pentane [BP $\propto$ VWf $\propto$ contact area $\propto \frac{1}{\text{Branching}}$]

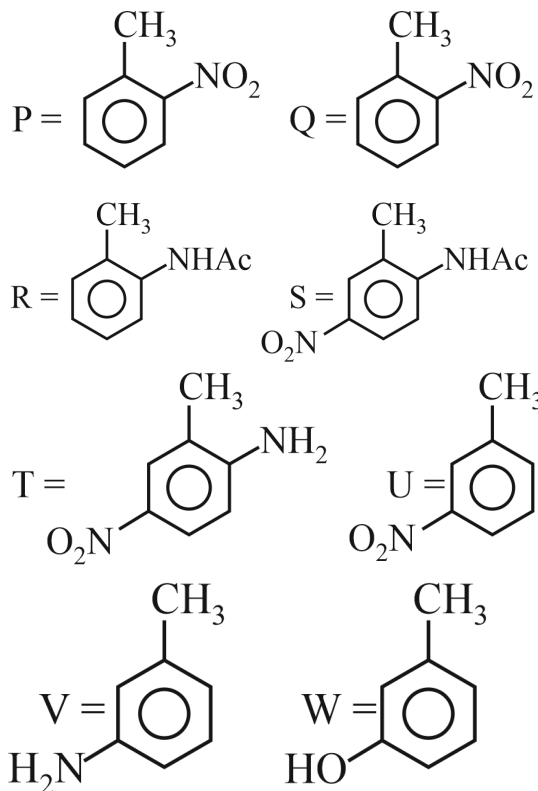
(C) Xe > Ar > He [BP $\propto$ mol. wt.]

(D)

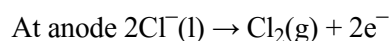
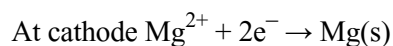
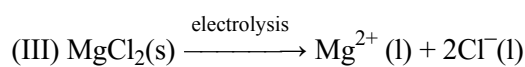
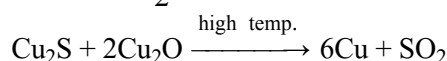
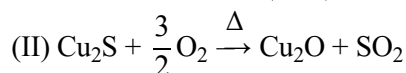


19. **Ans (B)**

In the given sequence of reactions the final product is Meta-methyl phenol



20. **Ans (C)**



PART-2 : CHEMISTRY

SECTION-II

1. **Ans (3)**

Number of hydroxyl group by acetylation

$$\text{method} = \frac{280 - \text{MW of } C_8H_{10}O_3}{42}$$

$$= \frac{280 - 154}{42} = \frac{126}{42} = 3$$

Number of phenolic hydroxyl

$$= \frac{\text{MW of methyl ether} - \text{MW}}{14}$$

$$= \frac{182 - 154}{14} = \frac{28}{14} = 2$$

Hence, $X = 2$

Number of alcoholic $-OH$ groups $= 3 - 2 = 1$

Hence, $Y = 1$

So, $X + Y = 2 + 1 = 3$.

2. **Ans (63)**

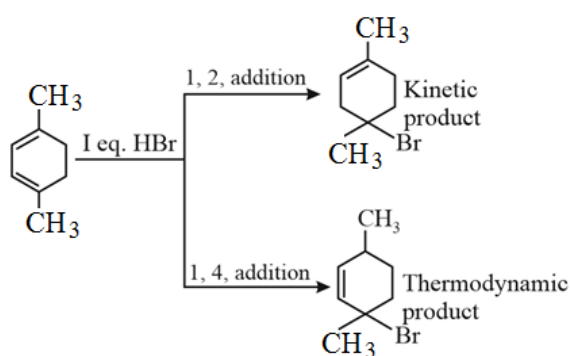
$$V \times 1.8 \times \frac{78.4}{100} = 0.9 \times 98$$

$$\therefore V = 62.50 \text{ ml}$$

3. **Ans (2)**

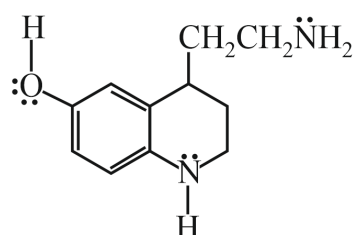
[M(EDTA)] type octahedral complex have two optically active isomers (one enantiomeric pair)

4. **Ans (12)**



5. **Ans (7)**

The structure of Serotonin is as follows

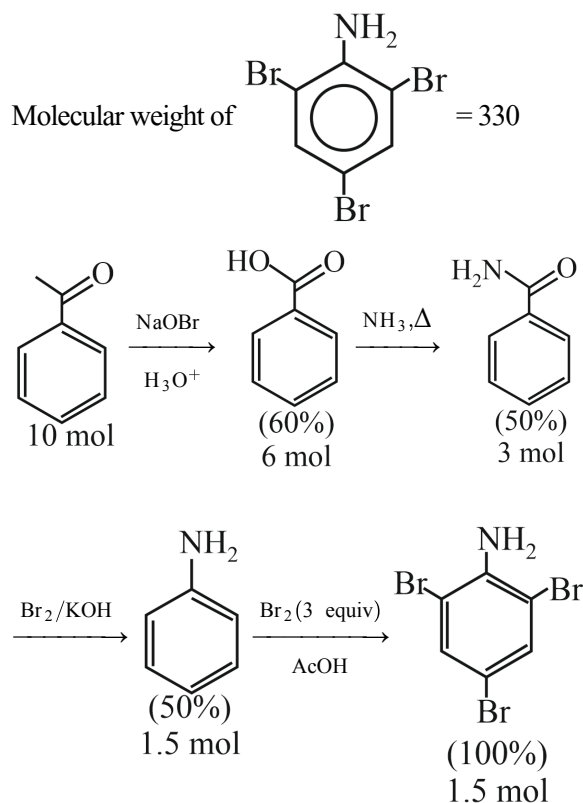


Number of lone pairs(X) $= 2 + 1 + 1 = 4$

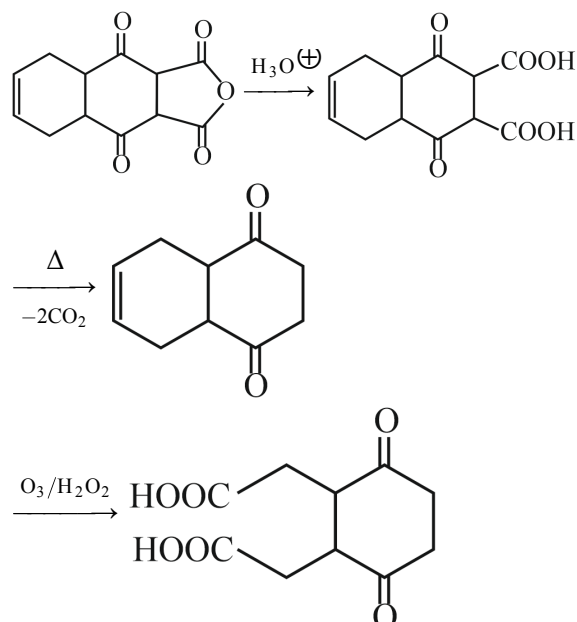
Number of π -bonds $= 3$

$$X + Y = 4 + 3 = 7$$

6. **Ans (495)**



7. **Ans (2)**



Hence in the final product number of groups is 2.

8. **Ans (5)**

$$P = 1 - \left(1 \times \frac{20}{100}\right) = 0.8 \text{ atm}$$

According to ideal gas equation, $PV = nRT$ or

$$PV = \frac{w}{M}RT,$$

$\therefore$ Weight of N_2 adsorbed by solid

$$= \frac{P \times V \times M}{R \times T}$$

$$= \frac{0.8 \times 1.5 \times 28}{0.08 \times 300}$$

$$= 1.4 \text{ g of } N_2$$

$$\therefore \frac{x}{m} = \frac{1.4 \text{ g}}{0.28 \text{ g}} = 5 \text{ g}$$

9. **Ans (7)**

(i), (iii), (vi), (vii), (ix), (xi) and (xiii)

10. **Ans (6)**

Adiabatic free expansion of ideal gas is also isothermal.

$$\Delta S = nR \ln \frac{V_2}{V_1} = 4 \times 2 \times \ln \left(\frac{20}{10}\right)$$

$$= 5.60 \text{ cal/K}$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (D)**

$$2\arg\left(\frac{z-i+3}{z+3i-1}\right) = \pi$$

$$\arg\left(\frac{z-i+3}{z+3i-1}\right) = \frac{\pi}{2}$$

$$(x+3)(x-1) + (y-1)(y+3) = 0$$

$$x^2 + 2x - 3 + y^2 + 2y - 3 = 0$$

$$x^2 + y^2 + 2x + 2y - 6 = 0$$

2. **Ans (A)**

$$\left(\int_0^{\pi/2} (\cos x)^{2022} dx\right) \left(\int_0^{\pi/2} (\cos x)^{2023} dx\right)$$

by wall's theorem

$$\frac{(2021)(2019)\dots\dots\dots 1}{(2022)(2020)\dots\dots\dots 2} \times \frac{(2022)\dots\dots\dots 2}{(2023)(2021)\dots\dots\dots 1}$$

$$\frac{1}{2023} \times \frac{\pi}{2}$$

3. **Ans (C)**

$$\lim_{n \rightarrow \infty} \frac{(1^{2022} + \dots\dots\dots n^{2022})n}{1^{2023} + 2^{2023} + \dots\dots\dots n^{2023}}$$

$$= \frac{2024}{2023}$$

4. **Ans (A)**

$$\left(\frac{\alpha\beta\gamma-2}{\gamma}\right) \left(\frac{\alpha\beta\gamma-2}{\alpha}\right) \left(\frac{\alpha\beta\gamma-2}{\beta}\right)$$

$$\frac{64}{6} = \frac{32}{3} = 10.66$$

5. **Ans (C)**

B B S R _ _ _ _ _

_ B B S R _ _ _ _ _

_ _ B B S R _ _ _ _ _

$$\text{Reqd. no of words} = 3 \times \frac{5!}{2} = 180$$

6. **Ans (A)**

$$S = 1 + 2 \left(\frac{10}{11}\right) + 3 \cdot \left(\frac{10}{11}\right)^2 + \dots\dots\dots \infty$$

$$(\text{Let } \frac{10}{11} = x)$$

$$S = 1 + 2x + 3x^2 + \dots\dots\dots \infty$$

$$S \cdot x = x + 2x^2 + \dots\dots\dots \infty$$

$$S(1-x) = 1 + x + x^2 + \dots\dots\dots \infty$$

$$S(1-x) = \frac{1}{1-x}$$

$$\Rightarrow S = \frac{1}{(1-x)^2} = \frac{1}{\left(1 - \frac{10}{11}\right)^2} = 121$$

7. **Ans (B)**

p	$\sim p$	$p \wedge \sim p$
T	F	F
T	F	F
F	T	F
F	T	F

$$\Rightarrow p \wedge (\sim p)$$

is contraction.

(B) $p \Rightarrow q \Leftrightarrow (\sim q \Rightarrow \sim p)$ is tautology, not a contradiction

(C) & (D) are obviously correct.

8. **Ans (B)**

$$\frac{x_1 + x_2 + x_3 + \dots\dots\dots + x_n}{n} = \bar{x}$$

New mean

$$= \frac{x_1 + x_2 + \dots\dots\dots + x_n}{n} + \frac{3 + 3^2 + \dots\dots\dots + 3^n}{n}$$

$$= \bar{x} + \frac{3 \cdot (3^n - 1)}{2n}$$

9. **Ans (A)**

$$y = mx + c$$

$$y' = m; y'' = 0; y''' = 0$$

$$0 + 0 + 2x \cdot m - mx - c + 1 = 0$$

$$\Rightarrow mx - c + 1 = 0$$

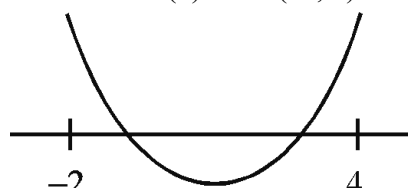
$$\Rightarrow m = 0; c = 1$$

10. **Ans (A)**

$$f(x) = 3x^2 - 3\alpha(2x) + 3(\alpha^2 - 1) = 3x^2 - 6\alpha x + 3(\alpha^2 - 1)$$

since, extrema lie in $(-2, 4)$

so roots of $f'(x)$ lie in $(-2, 4)$



$$f'(x) = 6x - 6\alpha = 0 \Rightarrow x = \alpha$$

$$\Rightarrow -2 < \alpha < 4$$

$$\Rightarrow -1 < \alpha < 3$$

Second method

$$\Rightarrow 12 + 12\alpha + 3\alpha^2 - 3 > 0$$

$$\Rightarrow 3\alpha^2 + 12\alpha + 9 > 0$$

$$\Rightarrow \alpha^2 + 4\alpha + 3 > 0,$$

$$\Rightarrow (\alpha + 1)(\alpha + 3) > 0$$

$$\Rightarrow \alpha \in (-\infty, -3) \cup (-1, \infty) \dots (1)$$

Also, $f(4) > 0$

$$\Rightarrow 48 - 24\alpha + 3\alpha^2 - 3 > 0$$

$$\Rightarrow 3\alpha^2 - 24\alpha + 45 > 0$$

$$\Rightarrow \alpha^2 - 8\alpha + 15 > 0$$

$$\Rightarrow (\alpha - 5)(\alpha - 3) > 0$$

$$\Rightarrow \alpha \in (-\infty, 3) \cup (5, \infty) \dots (2)$$

And, $D \geq 0$

$$\Rightarrow 36\alpha^2 - 4 \times 3 \times 3(\alpha^2 - 1) \geq 0$$

$$\Rightarrow \alpha^2 - \alpha^2 + 1 \geq 0 \text{ (this is always true)}$$

$$\text{Also } -2 < \frac{6\alpha}{2 \cdot 3} < 4 \Rightarrow \alpha \in (-2, 4) \dots (3)$$

Taking intersection of (1), (2), (3) $\alpha \in (-1, 3)$

11. **Ans (D)**

$$e^x(y + y') = 1$$

Integrating :-

$$ye^x = x + c \Rightarrow y = (x + c)e^{-x}$$

12. **Ans (D)**

$$3A + 2B = A^T \dots (1)$$

$$2A + 2B = A^T - A$$

$\therefore A^T - A$ is skew symmetric matrix

$$\therefore |2A + 2B| = 0$$

$$\Rightarrow |A + B| = 0$$

13. **Ans (D)**

$$\sum_{r=0}^{10} (-1)^r {}^{10}C_r \left(\frac{1}{3}\right)^r + \sum_{r=0}^{10} (-1)^r {}^{10}C_r \left(\frac{8}{9}\right)^r$$

$$\left(1 - \frac{1}{3}\right)^{10} + \left(1 - \frac{8}{9}\right)^{10}$$

$$\frac{2^{10}}{3^{10}} + \frac{1}{9^{10}}$$

$$\frac{6^{10} + 1}{3^{20}}$$

14. **Ans (B)**

Let the family of circles be

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

then condition for orthogonality

$$2g \left(2 \left(\lambda - \frac{3}{2}\right)\right) + 2f \left(\frac{3}{2} \left(\lambda - \frac{4}{3}\right)\right) = c - 6(\lambda + 2)$$

$$\lambda [4g + 3f + 6] - \left[6g - \frac{8f}{3} - c + 12\right] = 0$$

This is true for all $\lambda \in \mathbb{R}$

$$4g + 3f + 6 = 0 \text{ \& } 6g - \frac{8f}{3} - c + 12 = 0$$

Hence center lies on

$$-4x - 3y + 6 = 0$$

$$\Rightarrow 4x + 3y - 6 = 0$$

15. Ans (D)

Let $x^2 + x = t$

$x^2 + x - t = 0 \quad (\forall x \in \mathbb{R})$

$D = 1 + 4t \geq 0 \Rightarrow t \geq \frac{-1}{4} \Rightarrow \left[\frac{-1}{4}, \infty \right)$

So, quadratic, $t^2 + at + 4 = 0$; $t \in \left[\frac{-1}{4}, \infty \right)$

All four roots are real & distinct

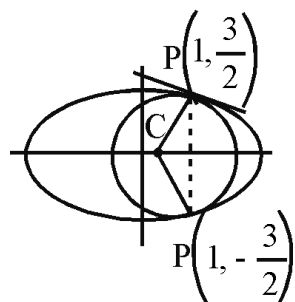
(i) $D > 0 \Rightarrow a \in (-\infty, -4) \cup (4, \infty)$

(ii) $f\left(\frac{-1}{4}\right) > 0 \Rightarrow a < \frac{65}{4}$

(iii) $\frac{-b}{2a} > -\frac{1}{4} \Rightarrow a < \frac{1}{2}$

$\therefore a \in (-\infty, -4)$

16. Ans (B)



by symmetry centre of circle lies on x-axis

Normal at P is $\frac{4x}{1} - \frac{3y}{3/2} = 1$

$\Rightarrow$ Point C is $\left(\frac{1}{4}, 0\right)$

Radius = $\sqrt{\left(1 - \frac{1}{4}\right)^2 + \left(\frac{3}{2}\right)^2} = \sqrt{\frac{9}{16} + \frac{9}{4}} = \frac{3\sqrt{5}}{4}$

17. Ans (B)

ways = $(5 + 1)(6 + 1)(3 + 1) - 1$
 $= 167$

18. Ans (C)

$\sum_{r=1}^9 \sin^2 \frac{r\pi}{10} = \frac{1}{2} \sum_{r=1}^9 \left(1 - \cos \frac{2r\pi}{10}\right) = 5$

19. Ans (B)

$\vec{v} = (\hat{a} \times \hat{b}) \times \hat{a} + (\hat{a} \times \hat{b}) \times \hat{b} + (\hat{a} \times \hat{b}) \times \hat{c}$
 $= \hat{b} - (\hat{a} \cdot \hat{b})\hat{a} + (\hat{a} - \hat{b})\hat{b} - \hat{a} + (\hat{a} \cdot \hat{c})\hat{b} - (\hat{b} \cdot \hat{c})\hat{a}$
 $= 2(\hat{b} - \hat{a})$

20. Ans (B)

$$\begin{aligned} & \int_0^{\pi/2} \frac{4\cos^4\theta}{\cos^4\theta + \sin^2 2\theta} d\theta \\ &= 4 \int_0^{\pi/2} \frac{\cos^2\theta}{\cos^2\theta + 4\sin^2\theta} d\theta \\ &= 4 \int_0^{\pi/2} \frac{\sec^2\theta}{1 + \tan^2\theta} \frac{d\theta}{(1 + 4\tan^2\theta)} \quad \tan\theta = x \\ &= \int_0^{\infty} \frac{4dx}{(1+x^2)(1+4x^2)} \\ &= 4 \times \frac{\pi}{6} = \frac{2\pi}{3} \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (14)

$f(x) = (x-1)(x^2 + ax + b) \mid x^2 + 8x + 6 \mid$

for $f(x)$ to be derivable in $(-\infty, \infty)$, both roots of both the equations $x^2 + ax + b = 0$ & $x^2 + 8x + 6 = 0$ must be common.

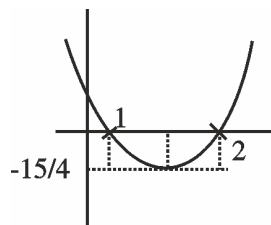
$\therefore a = 8$ & $b = 6$

2. Ans (25)

$g(x) = \sin^2 x + \cos^2 x - \frac{x^2}{2} + \frac{x^2}{2} + 4 = 5$

$y = 15(x^2 - 3x + 2)$

Reqd. area = $\frac{2}{3} \times \frac{15}{4} = \frac{5}{2} = 2.50$

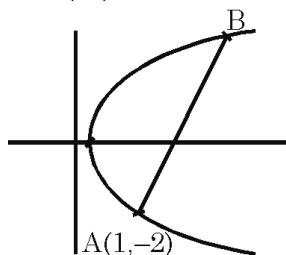


3. Ans (5)

Required probability = $\frac{{}^7C_2}{{}^7C_2 + {}^7C_1 \cdot {}^5C_1} = \frac{3}{8} \equiv \frac{p}{q}$

$\therefore |p - q| = 5$

4. Ans (4)



Parameter of A : $2 \times 1 \times t = -2$

$\therefore$ Parameter of B : $t_2 = -(-1) - \frac{2}{-1} = 3$

$\therefore B(9, 6)$

$\therefore AB = 8\sqrt{2}$

5. Ans (4)

The value of n is the least degree of numerator

$$\text{Nr.} = (\cos x - 1)(\cos x - e^x) - \frac{x^3}{2}$$

(expanding this)

$$= \left(1 - \frac{x^2}{2} + \frac{x^4}{24} \dots - 1\right)$$

$$\left(\left(1 - \frac{x^2}{2} + \frac{x^4}{24} \dots\right) - \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} \dots\right)\right) - \frac{x^3}{2}$$

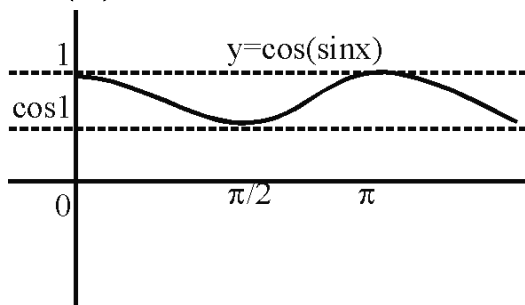
$$= \left(-\frac{x^2}{2} + \frac{x^4}{24} \dots\right) \left(-x - x^2 - \frac{x^3}{6} - \frac{x^5}{120} \dots\right) - \frac{x^3}{2}$$

$$= \frac{x^3}{2} - \frac{x^5}{24} \dots + \frac{x^4}{2} - \frac{x^6}{24} \dots - \frac{x^3}{2}$$

So, minimum degree of numerator is 4

So, the value of n is 4.

6. Ans (2)



7. Ans (64)

Normal at $\left(ct, \frac{c}{t}\right)$ intersect the curve again

$\left(ct', \frac{c}{t'}\right)$ then $t't^3 = -1$.

$$\Rightarrow (x', y') = \left(-\frac{1}{4}, -16\right)$$

8. Ans (3)

$$\tan(\theta + (60 - \theta)) = \sqrt{3}$$

$$\Rightarrow \tan \theta + \tan(60 - \theta) = \sqrt{3} - \sqrt{3} \tan \theta \tan(60 - \theta)$$

$$\Rightarrow \tan \theta + \tan(60 - \theta) + \sqrt{3} \tan \theta \tan(60 - \theta) = \sqrt{3}$$

$$\Rightarrow \tan \theta + \sqrt{3} \tan^2 \theta + \tan(60 - \theta) (1 + \sqrt{3} \tan \theta)$$

$$= \sqrt{3} \sec^2 \theta$$

$$\Rightarrow (1 + \sqrt{3} \tan \theta)(\tan \theta + \tan(60 - \theta)) = \sqrt{3} \sec^2 \theta$$

$$\Rightarrow \frac{(1 + \sqrt{3} \tan \theta)(\tan \theta + \tan(60 - \theta))}{(1 + \tan^2 \theta)} = \sqrt{3}$$

Now, put $\theta = 1^\circ$ and $\theta = 2^\circ$

$$\frac{(1 + \sqrt{3} \tan 1^\circ)(1 + \sqrt{3} \tan 2^\circ)(\tan 10^\circ + \tan 59^\circ)(\tan 2^\circ + \tan 54^\circ)}{(1 + \tan^2 1^\circ)(1 + \tan^2 2^\circ)}$$

$$= \sqrt{3} \cdot \sqrt{3} = 3$$

9. Ans (1)

$$\sum_{r=1}^{19} \frac{3(r+1)r+1}{(r+1)(r^5+2r^4+r^3)}$$

$$= \sum_{r=1}^{19} \frac{1}{r^3} - \frac{1}{(r+1)^3}$$

$$T_r = \frac{1}{r^3} - \frac{1}{(r+1)^3}$$

$$S = 1 - \frac{1}{20^3} = \frac{8000-1}{8000} = \frac{7999}{8000}$$

10. Ans (8)

$$I = \int_0^9 e^x (1 + \{x\} + \{x\}^2) dx = \int_0^1 + \int_1^2 + \int_2^3 + \dots + \int_8^9$$

$$\text{in integration } \sum_{k=0}^8 \int_k^{k+1} e^x (1 + \{x\} + \{x\}^2) dx$$

Put $x = k + t$

$$I = \sum_{k=0}^8 \int_0^1 e^k \cdot e^t (1 + \{t\} + \{t\}^2) dt$$

$$I = \sum_{k=0}^8 e^k \cdot J$$

$$I = J (e^0 + e^1 + e^2 + \dots + e^8)$$

$$\Rightarrow \sum_{k=1}^8 e^k \Rightarrow k = 8$$

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B	A,D	A,C	A,C	A,C	A,B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	A	B	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	6.00	8.00	5.00	6.28	4.00	3.14	2.00

PART-2 : CHEMISTRY

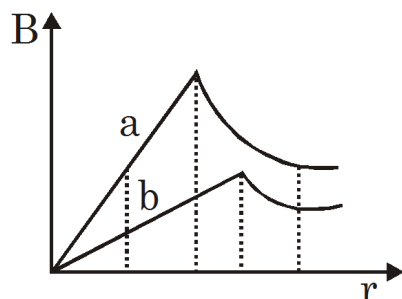
SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	A	B,C	A,B,D	B,C	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	C	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	37.00	4.00	57.50	4.00	300.00	6.00	5.00	4.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,D	C	B,C,D	B	A,B,C,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	B	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	18.00	1.00	0.00	81.00	3.00	3.00	3.00	750.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (B)



$$B = \frac{\mu_0 i}{2\pi r}; r > R$$

$$B = \frac{\mu_0 j r}{2}; r < R$$

$$\frac{\mu_0 j_a r}{2} > \frac{\mu_0 j_b r}{2}; r > R$$

$$j_a > j_b$$

$$\frac{\mu_0 i_a}{2\pi r} > \frac{\mu_0 i_b}{2\pi r}; i_a > i_b, r > R$$

$$\frac{\mu_0 i_b}{2\pi b} < \frac{\mu_0 i_a}{2\pi a}; \frac{i_b}{b} < \frac{i_a}{a}$$

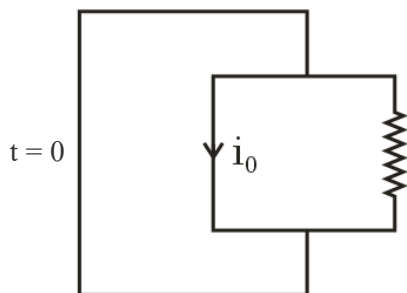
2. Ans (A,D)

$$\tau = i \left[\frac{\pi r^2}{4} - \frac{r^2}{4} \right] B_x$$

$$\tau = \frac{i r^2}{4} (\pi - 1) B_x$$

force on loop in constant field is zero.

3. Ans (A,C)

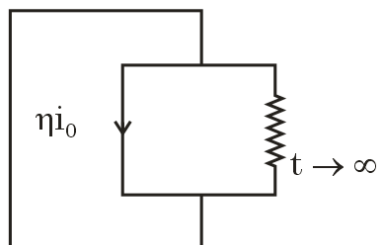


$$\frac{L di}{dt} = iR$$

$$L di = dqR$$

$$\frac{L}{R} \int_{I_0}^{\eta I_0} di = q$$

$$\frac{L}{R} (\eta - 1) I_0 = q$$



$$P = Vi$$

$$P = \frac{L di}{dt} i$$

$$\int P dt = L \int i di$$

$$\text{Heat} = \frac{L}{2} [i^2]_0^{(\eta-1)I_0}$$

$$\text{Heat} = \frac{1}{2} L I_0^2 (\eta - 1)^2$$

4. Ans (A,C)

When B and D is cut

$$mg - T = ma$$

taking torque about mid point of rod

$$T \cdot \frac{\ell}{2} = \frac{m\ell^2}{12} \times \frac{a}{\ell/2}$$

$$T = mg/4$$

5. Ans (A,C)

$$h\nu = \phi + eV_0$$

$$V_0 = \frac{h\nu}{e} - \frac{\phi}{e}$$

$$V_0 = \frac{hc}{e} \cdot \frac{1}{\lambda} - \frac{\phi}{e}$$

Here intercept on y-axis is $\frac{\phi}{e}$.

6. Ans (A,B,D)

$$\lambda_A = vT + v_S T$$

$$\lambda_A = \frac{7v}{4f}$$

$$\lambda_B = \frac{7v/4f}{1/5} = \frac{35v}{4f}$$

$$f_A = \frac{v}{7v/4f} = \frac{4f}{7}$$

$$f_B = \frac{\frac{14}{3}v}{\frac{35v}{4f}}$$

$$f_B = \frac{8f}{15}$$

PART-1 : PHYSICS

SECTION-I (ii)

8. Ans (A)

$$\Delta Q = v_i - v_f$$

$$\Delta Q = \frac{Q_i^2}{2C} - \frac{Q_f^2}{2C}$$

$$\Delta Q = \frac{1}{2} \times 75 \times 10^{-3} \times \frac{640}{3} \times \frac{640}{3}$$

$$\Delta Q = \frac{1}{2} \times \frac{4 \times 4}{75 \times 10^{-3}}$$

$$Q_f = Q_0 e^{-\frac{t}{RC}}$$

$$Q_f = \frac{640}{3} \times 75 \times 10^{-3} \times e^{-\frac{2.5 \ln 4}{150}}$$

$$Q_f = \frac{640}{3} \times 75 \times 10^{-3} \times \frac{1}{4}$$

$$Q_f = 4C$$

$$\Delta Q = 1706 - 106$$

$$\Delta Q = 1600 = 1.6 \text{ kJ}$$

$$\Delta Q + \Delta w = 1.6 \text{ kJ}$$

$$\Delta u + nR\Delta T = 1.6 \text{ kJ}$$

$$\Delta u = 1.6 \text{ kJ} - \frac{1 \times 8.314 \times 72}{1000} \text{ kJ}$$

$$\Delta u = 1.6 \text{ kJ} - 0.6 \text{ kJ} = 1 \text{ kJ}$$

$$w = 0.6 \text{ kJ}$$

9. Ans (B)

$$P_0 + \frac{F}{A} = P_0 + \frac{1}{2} \rho v^2$$

$$\sqrt{\frac{2F}{A\rho}} = v$$

10. Ans (D)

$$v_x = \sqrt{\frac{2F}{A\rho}}, v_y = \sqrt{2gh}$$

$$v = \sqrt{\frac{2F}{A\rho} + 2gh}$$

$$v = \sqrt{\frac{2F + 2ghA\rho}{A\rho}}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (5.00)

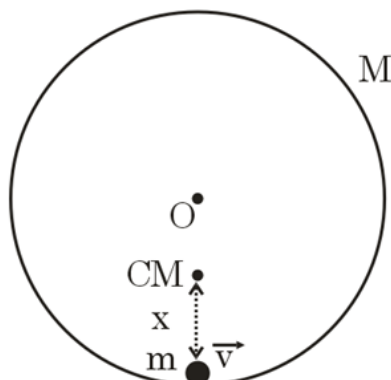
For central bright fringe C

$$\left(\frac{4}{3} \times 2\right) + \left(\frac{5}{3} \times 1\right) + \frac{\mu}{3}(D - 1)$$

$$= 2 + \left(\frac{8}{3} \times \frac{3}{2}\right) + \frac{\mu}{3}\left(D - \frac{3}{2}\right)$$

$$\mu = 5$$

2. Ans (6.00)



$$v_{CM} = \frac{mv}{m + M}; x = \frac{MR}{M + m}$$

$$N = \frac{m(v - v_{cm})^2}{x} = \frac{mv^2 M}{(m + M)R} = 6 \times 10^{-3} \text{ N}$$

3. Ans (8.00)

$$\frac{dN_x}{dt} = \frac{dN_y}{dt}$$

$$\lambda_x N_x = \lambda_y N_y$$

$$\frac{\ln 2}{1} (8N_0 e^{-\lambda_x t}) = \frac{\ln 2}{2} (N_0 e^{-\lambda_y t})$$

$$t = \frac{4 \ln 2}{\lambda_x - \lambda_y} = \frac{4 t_x t_y}{t_y - t_x} = 8$$

4. Ans (5.00)

From conservation of energy

$$(2M \cdot g \cdot 3R) - (M \cdot g \cdot 3R)$$

$$= \frac{1}{2} \left\{ \frac{M(6R)^2}{12} + M(3R)^2 + (2M \{3R\})^2 \right\} \omega^2$$

$$= \frac{1}{2} (30MR^2) \omega^2 = 15 (MR^2) \omega^2$$

$$3MgR = 15 MR^2 \omega^2$$

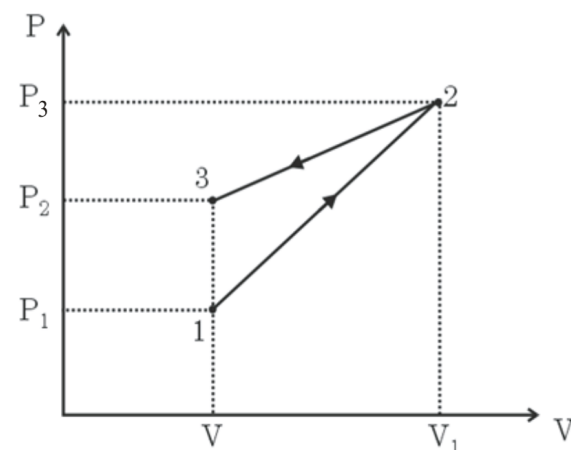
$$\Rightarrow \omega = \sqrt{\frac{g}{5R}}$$

5. Ans (6.28)

$$\pi(2)^2 i + \sqrt{\pi} \sqrt{\pi} \hat{i} = 5\pi i$$

$$= 5 \times 3.14 \times \frac{2}{5} = 6.28$$

6. Ans (4.00)



$$\Delta Q = \Delta u + \Delta w$$

$$0 = \frac{nf}{2} (P_2 V - P_1 V)$$

$$+ \left[\frac{1}{2} (P_1 + P_3) (V_1 - V) - \frac{1}{2} (P_2 + P_3) (V_1 - V) \right]$$

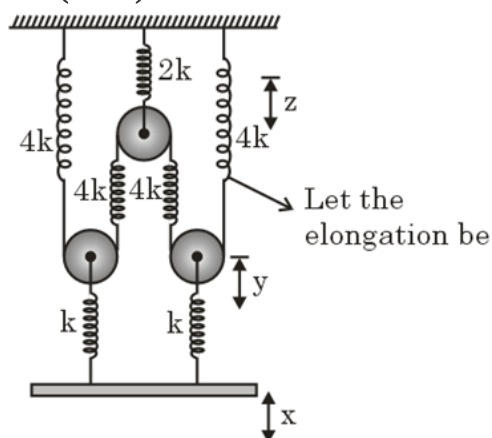
$$(P_2 - P_1) \left(\frac{3V}{2} - \frac{V_1 - V}{2} \right) = 0$$

$$V_1 = 4V$$

7. Ans (3.14)

$$\begin{aligned} \frac{1}{2}k(x+x_0)^2 - mgx + \frac{1}{2}\epsilon_0 B^2 v^2 \\ + \frac{1}{2}mv^2 = \text{constant} \\ \frac{2}{2}k(x+x_0)v - mgv + \frac{1}{2}\epsilon_0 B^2 v 2V a \\ + \frac{1}{2}m 2va = 0 \\ kx + \epsilon_0 B^2 Va + ma = 0 \\ (\epsilon_0 B^2 V + m)a = -kx \\ a = \frac{-k}{(\epsilon_0 B^2 V + m)} \\ T = 2\pi\sqrt{\frac{m + \epsilon_0 B^2 V}{k}} \\ T = 3.14 \text{ sec} \end{aligned}$$

8. Ans (2.00)



$$\begin{aligned} 2kz &= 4kz + 4kz + k(x-y) \\ 8kz &= 2kz + 4kz + k(x-y) \\ 4kz &= k(x-y) \quad \dots (i) \\ 4kz &= 4y - 2z \quad \dots (ii) \\ 3z &= 4y \\ T &= 2k(x-y) \quad \dots (iii) \\ T &= 2k\left(x - \frac{32}{11}\right) \\ T &= \frac{16}{11}kx \\ 2k(x-y) &= 16k \\ x-y &= 8 \\ x-y &= 2z \\ x &= 22 + \frac{32}{4}; x = \frac{11z}{4} \\ x &= \frac{11}{4} \cdot \frac{4y}{3}; x = \frac{11}{3}y \end{aligned}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A,B,C)

Cell reaction is :



(saturated sol) + 2Ag(s)

$$\begin{aligned} \Delta G_{0^\circ\text{C}}^\circ &= -nFE_{\text{cell}}^\circ = -2 \times 96500 \times 0.70 \\ &= -135100 \text{ J} \end{aligned}$$

$$\begin{aligned} \Delta G_{50^\circ\text{C}}^\circ &= -nFE_{\text{cell}}^\circ = -2 \times 96500 \times 0.60 \\ &= -115800 \text{ J} \end{aligned}$$

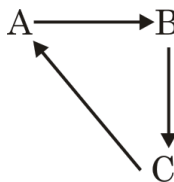
$$\begin{aligned} \Delta S^\circ &= nF \left(\frac{\Delta E}{\Delta T} \right) \\ &= 2 \times 96500 \times \left(\frac{0.60 - 0.70}{328 - 273} \right) \\ &= -386 \text{ J/K} \end{aligned}$$

$$\Delta G^\circ = \Delta H^\circ - T \cdot \Delta S^\circ$$

$$\begin{aligned} \therefore \Delta H^\circ &= \Delta G^\circ + T \cdot \Delta S^\circ \\ &= (-135100) + 273 \times (-386) \\ &= -240478 \text{ J} \end{aligned}$$

2. Ans (A)

Over all process is cyclic



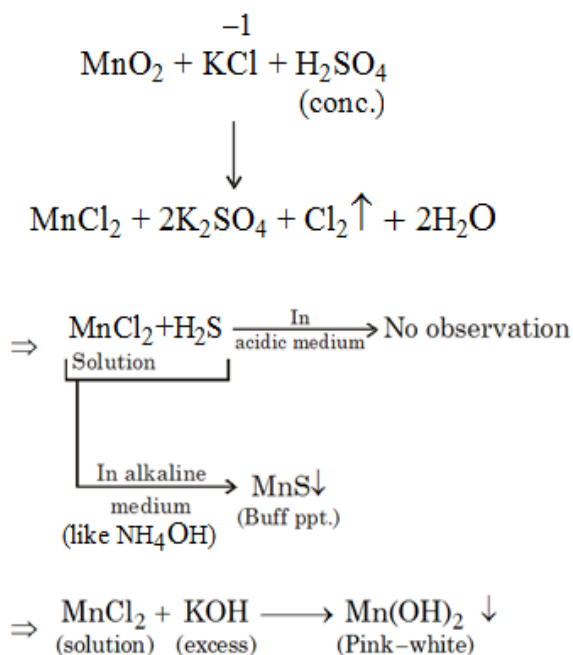
Thus, $\Delta E = 0$, $\Delta H = 0$, $\Delta S = 0$, $\Delta E = q + W$

$$0 = q + W \Rightarrow q = -W$$

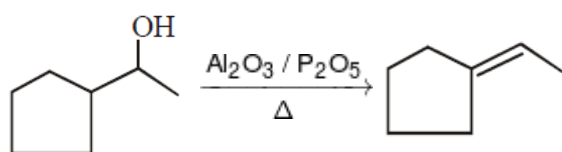
|Total work| = Area enclosed

$$\begin{aligned} &= \frac{1}{2} \times 20 \times 0.5 \\ &= 5 \text{ L-atm} = 506.5 \text{ J} \end{aligned}$$

4. Ans (A,B,D)



5. Ans (B,C)



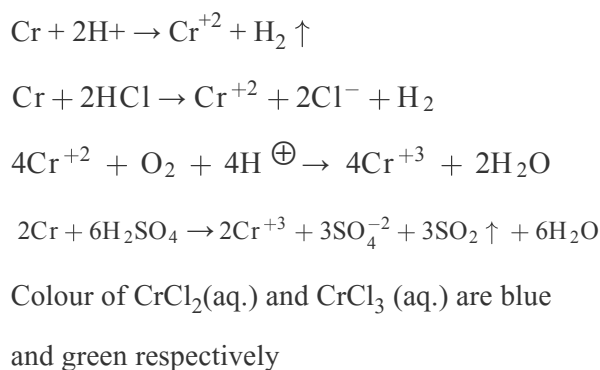
As no rearrangement takes place therefore possibility of conc. H_2SO_4 & conc. H_3PO_4 is neglected.

Above reaction follows E_2 Mechanism.

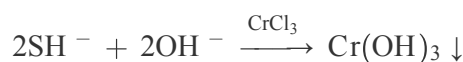
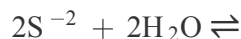
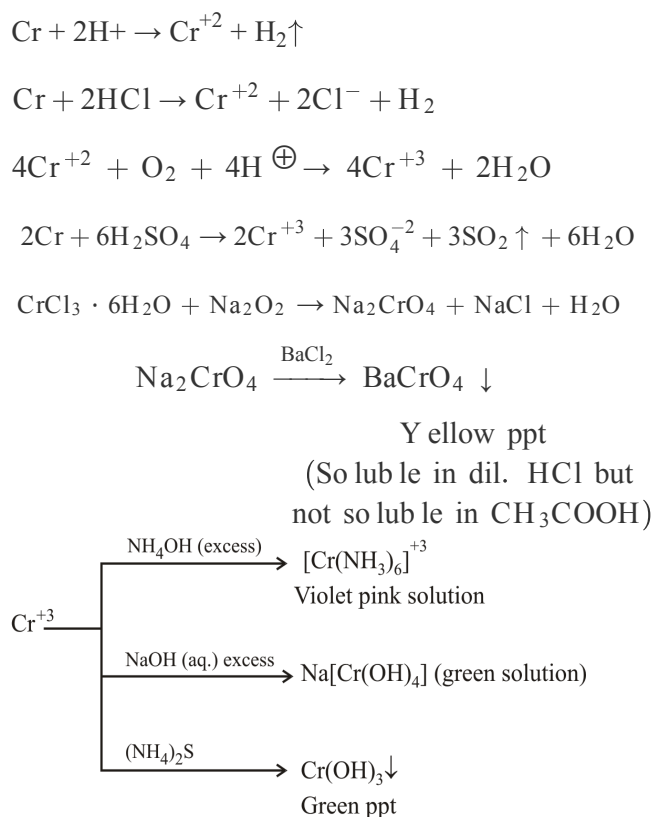
PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (A)



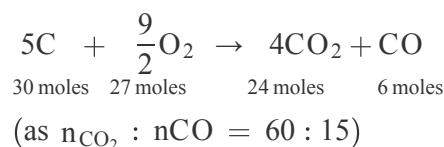
8. Ans (C)



PART-2 : CHEMISTRY

SECTION-II

1. Ans (37.00)



$$X = 27 + 6 \times \frac{25}{15} = 37 \text{ moles}$$

2. Ans (4.00)

$$P \propto T^3 \Rightarrow pV^{1.5} = \text{constant}$$

$$\therefore C_{V,m} = \frac{R}{1.5 - 1.0} = 2R$$

$$\begin{array}{l} \text{Now, } q_V = n.C_{V,m} \cdot \Delta T \\ = 1 \times 2R \times (2000 - 1000) \\ = 4 \text{ kcal} \end{array}$$

3. **Ans (57.50)**

$$\Delta T_b = K_b \cdot m$$

$$\text{Molality} = \frac{1.8}{\frac{M_{\text{solute}}}{90 \times 10^{-3}}} \quad M_{\text{solute}} = \text{Molar mass of solute g/mol} \rightarrow \text{Mass of solvent in kg}$$

$$\Delta T_b = 354.11 - 353.23 = 0.88$$

$$\Delta T_b = 0.88 = (2.53) \times \frac{1.8}{M_{\text{solute}} \times 90 \times 10^{-3}}$$

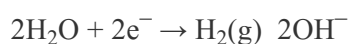
$$M_{\text{solute}} = 57.5 \text{ g/mole}$$

4. **Ans (4.00)**

$$[H^+] = 10^{-9} M$$

$$[OH^-] = 10^{-5} M$$

$$n_{OH^-} = 10^{-5} \times 0.8$$



$$\frac{0.193 \times t}{96500} \times 8 \times 10^{-6}$$

$$t = 4 \text{ sec.}$$

5. **Ans (300.00)**

MO can be reduced by C at 2000°C.

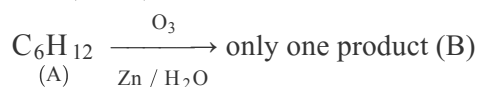


$$|\Delta G_r| = 300 \text{ kJ/mole}$$

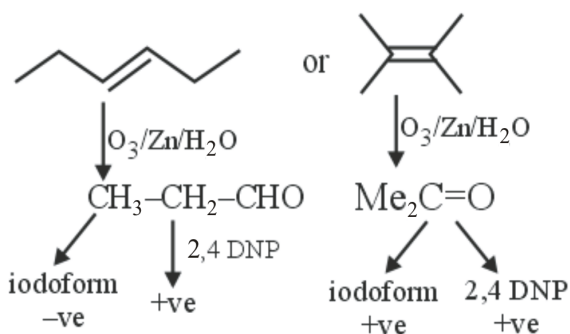
6. **Ans (6.00)**

Except reaction (d) all produces O_2

8. **Ans (4.00)**



Thus (A) is a symmetrical alkene.



Hence it is hex-3-ene.

PART-3 : MATHEMATICS

SECTION-I (i)

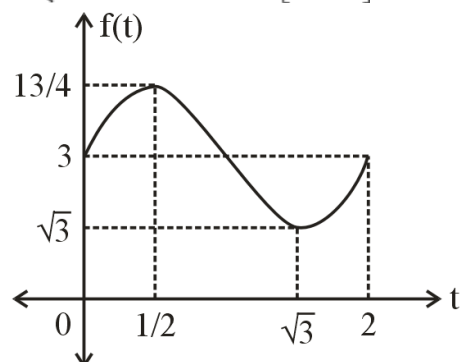
1. **Ans (A,D)**

$$\text{Let } t = |z + 1|, t \in [0, 2]$$

$$\therefore t^2 = (1+z)(1+\bar{z}) = 2 + 2 \operatorname{Re}(z) \Rightarrow \operatorname{Re}(z) = \frac{t^2 - 2}{2}$$

$$|1-z+z^2|^2 = (1-z+z^2)(1-\bar{z}+\bar{z}^2) = (1-z+z^2) \left(1 - \frac{1}{z} + \frac{1}{z^2}\right) = \left(\frac{z^2 - z + 1}{z}\right)^2 = (z + \bar{z} - 1)^2 = (2 \operatorname{Re}(z) - 1)^2 = (t^2 - 3)^2$$

$$\therefore f: [0, 2] \rightarrow \mathbb{R}, f(t) = t + |t^2 - 3| = \begin{cases} -t^2 + t + 3, & t \in [0, \sqrt{3}] \\ t^2 + t - 3, & t \in [\sqrt{3}, 2] \end{cases}$$



$$\therefore \sqrt{3} \leq t + |t^2 - 3| \leq \frac{13}{4}$$

2. **Ans (C)**

$$f^5 f' + g^2 f f' + f^2 g g' + g^3 g' = 0$$

$$f^5 f' + g f (g f' + f g') + g^3 g' = 0$$

$$f^5 f' + g f (g f)' + g^3 g' = 0$$

$$\frac{f^6}{6} + \frac{(g f)^2}{2} + \frac{g^4}{4} = c$$

$\therefore f$ & g both are bounded

3. Ans (B,C,D)

$$I_1 = \int_0^{\pi/2024} \sin(2\pi \cos^2(1012x)) dx$$

$$I_1 = \int_0^{\pi/2024} \sin(2\pi \sin^2(1012x)) dx$$

$$\therefore 2I_1 = \int_0^{\pi/2024} 2 \sin \pi \cdot \cos(\pi \cos(2024x)) dx = 0$$

$$\Rightarrow I_1 = 0$$

$$I_2 = \int_0^{\pi/2024} \sin(\pi \cos^2(1012x)) dx$$

$$I_2 = \int_0^{\pi/2024} \sin(\pi \sin^2(1012x)) dx$$

$$\therefore 2I_2 = \int_0^{\pi/2024} 2 \cos \frac{(\pi \cos(2024x))}{2} dx$$

$$I_2 = 2 \int_0^{\pi/4048} \cos\left(\frac{\pi}{2} \cos(2024x)\right) dx$$

$$\text{Let } 2x = t \Rightarrow 2dx = dt$$

$$I_2 = \int_0^{\pi/2024} \cos\left(\frac{\pi}{2} \cos(1012t)\right) dt$$

$$= \int_0^{\pi/2024} \cos\left(\frac{\pi}{2} (1 - 2\sin^2(506t))\right) dt$$

$$= \int_0^{\pi/2024} \sin(\pi \sin^2(506t)) dt$$

$$= I_3$$

4. Ans (B)

$$|\vec{r} - \vec{b}| + |\vec{r} - \vec{c}| = 4 \text{ represent an ellipse in 3D}$$

$$\text{space, with } a = 2, |\vec{b} - \vec{c}| = 2ae$$

$$\therefore b = \frac{1}{\sqrt{2}},$$

$$\vec{r} \cdot \vec{a} = 5 \text{ is plane passing through } \vec{b}, \vec{c}$$

$$\text{hence area} = \pi ab = \pi(2) \frac{1}{\sqrt{2}} = \sqrt{2}\pi \text{ sq. unit}$$

5. Ans (A,B,C,D)

$$P + Q = PQ \quad PQ = P^2 + P \Rightarrow Q = P^2$$

$$QPQ = P^5 = P + I \Rightarrow P(P^4 - I) = I$$

$$Q^5 - P^5 = (P^5)^2 - P^5 = (P + I)^2 - (P + I)$$

$$\Rightarrow P^2 + P = P + Q \Rightarrow P^2 = Q$$

6. Ans (A,B,C,D)

All options are not always correct

for option (C) :

$$f(x) = \begin{cases} x + x^2 \sin\left(\frac{2}{x}\right) & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

The equation of the given ellipse is

$$x^2/a^2 + y^2/b^2 = 1 \quad \dots(1)$$

Let the equation of any confocal to the ellipse (1) be

$$x^2/(a^2 + \lambda) + y^2/(b^2 + \lambda) = 1$$

If it passes through the point (x_1, y_1) , we get

$$x_1^2/(a^2 + \lambda) + y_1^2/(b^2 + \lambda) = 1$$

$$\Rightarrow (b^2 + \lambda)x_1^2 + (a^2 + \lambda)y_1^2 = (a^2 + \lambda)(b^2 + \lambda) = 0 \quad \dots(2)$$

$$\lambda^2 + \lambda(a^2 + b^2 - x_1^2 - y_1^2) + (a^2b^2 - b^2x_1^2 - a^2y_1^2) = 0$$

The equation (2) is a quadratic equation in λ and

according to the question the two values of λ are given to

be λ_1 and λ_2 .

We have

$$\lambda_1 + \lambda_2 = -(a^2 + b^2 - x_1^2 - y_1^2)$$

$$\therefore x_1^2 + y_1^2 - a^2 - b^2 = \lambda_1 + \lambda_2 \quad \dots(3)$$

$$\text{Also } \lambda_1 \lambda_2 = (a^2b^2 - b^2x_1^2 - a^2y_1^2)$$

8. **Ans (B)**

The equation of the given ellipse is

$$x^2/a^2 + y^2/b^2 = 1 \quad \dots(1)$$

Let the equation of any confocal to the ellipse (1) be

$$x^2/(a^2 + \lambda) + y^2/(b^2 + \lambda) = 1$$

If it passes through the point (x_1, y_1) , we get

$$x_1^2/(a^2 + \lambda) + y_1^2/(b^2 + \lambda) = 1$$

$$\Rightarrow (b^2 + \lambda)x_1^2 + (a^2 + \lambda)y_1^2 = (a^2 + \lambda)(b^2 + \lambda) = 0 \quad \dots(2)$$

$$\lambda^2 + \lambda(a^2 + b^2 - x_1^2 - y_1^2) + (a^2b^2 - b^2x_1^2 - a^2y_1^2) = 0$$

The equation (2) is a quadratic equation in λ and according to the question the two values of λ are given to be λ_1 and λ_2 .

We have

$$\lambda_1 + \lambda_2 = -(a^2 + b^2 - x_1^2 - y_1^2)$$

$$\therefore x_1^2 + y_1^2 - a^2 - b^2 = \lambda_1 + \lambda_2 \quad \dots(3)$$

$$\text{Also } \lambda_1 \lambda_2 = (a^2b^2 - b^2x_1^2 - a^2y_1^2)$$

9. **Ans (B)**

$$P_n = 4^n + {}^nC_2 \cdot 4^{n-2} + {}^nC_4 (4)^{n-4} + \dots$$

$$Q_n = {}^nC_1 4^{n-1} + {}^nC_3 4^{n-3} + \dots$$

$$P_n + Q_n = (1 + 4)^n = 5^n$$

$$P_n - Q_n = (4 - 1)^n = 3^n$$

$$\therefore P_n = \frac{5^n + 3^n}{2}, Q_n = \frac{5^n - 3^n}{2}$$

10. **Ans (A)**

$$P_n = 4^n + {}^nC_2 \cdot 4^{n-2} + {}^nC_4 (4)^{n-4} + \dots$$

$$Q_n = {}^nC_1 4^{n-1} + {}^nC_3 4^{n-3} + \dots$$

$$P_n + Q_n = (1 + 4)^n = 5^n$$

$$P_n - Q_n = (4 - 1)^n = 3^n$$

$$\therefore P_n = \frac{5^n + 3^n}{2}, Q_n = \frac{5^n - 3^n}{2}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (18.00)**

Possibilities : babababa, ababbaba, abbababa, abababba, abababab

$$\Rightarrow 5 \left({}^{12}C_8 \cdot \frac{4}{2 \cdot 2} \right) = 14850$$

2. **Ans (1.00)**

Domain of $f(x) = x \in (-1, 1)$

$$\text{Put } \sin^{-1}x = t; t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$f(t) = 4\left(t - \frac{\pi}{4}\right)^2 + \pi^2$$

$$\therefore f_{\min.} \left(t = \frac{\pi}{4}\right) = \pi^2$$

3. **Ans (0.00)**

$$\text{tr}(AB(B'A' - A'B') + BA(A'B' - B'A')) = 0$$

$$\text{tr}((AB - BA)(B'A' - A'B')) = 0$$

$$\text{tr} \left(\underbrace{(AB - BA)}_X \underbrace{(AB - BA)'}_{X'} \right) = 0$$

$$\therefore X = 0$$

$$AB - BA = 0$$

$$\therefore |AB - BA| = 0$$

4. **Ans (81.00)**

Let removed numbers are $2r - 1, 2r + 3, 2r + 7$

$$\frac{n^2 - 3(2r + 3)}{n - 3} = \frac{105}{2}$$

$$\Rightarrow r = \frac{2n^2 - 105n + 297}{12}$$

$$\therefore 1 \leq \frac{2n^2 - 105n + 297}{12} \leq n - 4$$

$$\Rightarrow 50 \leq n \leq 55$$

Possible values of n are 51, 53, 55

For $n = 51, r = 12$

$\therefore$ Removed numbers are 23, 27, 31

5. **Ans (3.00)**

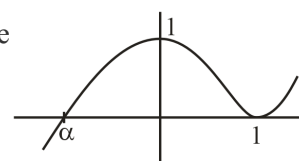
Roots of the $f(x) = 0$ are $\alpha \in (-1, 0)$ and 1

then $f(f(x)) = 0$ (where

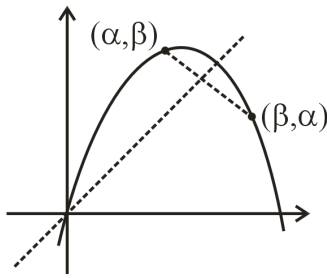
$$f(x) = t)$$

$$f(t) = 0 \Rightarrow t = \alpha, 1$$

So; total number of distinct real solutions = $2 + 1 = 3$



6. Ans (3.00)



(α, β) and (β, α) will lie on some line $y = -x + \lambda$

Solving line and parabola $-(x + \lambda) = ax - bx^2$

$$\Rightarrow bx^2 - (1 + a)x + \lambda = 0 \quad \dots(1)$$

Again, $y = a(\lambda - y) - b(\lambda - y)^2$

$$\Rightarrow by^2 + y(1 + a - 2b\lambda) + (b\lambda^2 - a\lambda) = 0 \quad \dots(2)$$

Both (1) and (2) has same roots which are α and β

$$\text{Hence, } 1 + a - 2b\lambda = -(1 + a) \Rightarrow \lambda = \frac{1 + a}{b}$$

Now, for α, β to exist disc of (1) > 0

$$\Rightarrow (1 + a)^2 - 4b\lambda > 0$$

$$\Rightarrow (a - 3)(a + 1) > 0$$

$$\Rightarrow a \in (-\infty, -1) \cup (3, \infty)$$

7. Ans (3.00)

$$\sin^2 x_1 + \sin^2 x_2 + \dots + \sin^2 x_{10} = 1$$

$$\cos x_i = \sqrt{\sum_{j=1}^{10} \sin^2 x_j}$$

For each $1 \leq i \leq 10$ we have

$$\cos x_i = \sqrt{\sum_{j=1}^{10} \sin^2 x_j} \geq \frac{\sum_{j \neq i} \sin x_j}{3}$$

$$\Rightarrow \sum_{i=1}^{10} \cos x_i \geq \sum_{i=1}^{10} \sum_{j \neq i} \frac{\sin x_j}{3} = \sum_{i=1}^{10} 9 \cdot \frac{\sin x_i}{3}$$

$$\Rightarrow \frac{\cos x_1 + \cos x_2 + \dots + \cos x_{10}}{\sin x_1 + \sin x_2 + \dots + \sin x_{10}} \geq 3$$

8. Ans (750.00)

$$\text{total number formed} = 7 \times 7 \times 6 \times 5 = 1470$$

$$\text{If 1 is not included} = 6 \times 6 \times 5 \times 4 = 720$$

$$\Rightarrow 1470 - 720 = 750$$